Is Physiological Equivalent Temperature (PET) a superior screening tool for heat stress risk than Wet-Bulb Globe Temperature (WBGT) index? Eight years of data from the Gothenburg half marathon

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ABSTRACT

Background The Wet-Bulb Globe Temperature (WBGT) index is a common tool to screen for heat stress for sporting events. However, the index has a number of limitations. Rational indices, such as the physiological equivalent temperature (PET) and Universal Thermal Climate Index (UTCI), are potential alternatives.

Aim To identify the thermal index that best predicts ambulance-required assistances and collapses during a city half marathon.

Methods Eight years (2010–2017) of meteorological and ambulance transport data, including medical records, from Gothenburg's half-marathon were used to analyse associations between WBGT, PET and UTCI and the rates of ambulance-required assistances and collapses. All associations were evaluated by Monte-Carlo simulations and leave-one-out-cross-validation.

Results The PET index showed the strongest correlation with both the rate of ambulance-required assistances (R^2=0.72, p=0.008) and collapses (R^2=0.71, p=0.008), followed by the UTCI (R^2=0.64, p=0.017; R^2=0.64, p=0.017) whereas the WBGT index showed substantially poorer correlations (R^2=0.56, p=0.031; R^2=0.56, p=0.033). PET stages of stress, match the rates of collapses better that the WBGT flag colour warning. Compared with the PET, the WBGT underestimates heat stress, especially at high radiant heat load. The rate of collapses increases with increasing heat stress; large increase from the day before the race seems to have an impact of the rate of collapses.

Conclusion We contend that the PET is a better predictor of collapses during a half marathon than the WBGT. We call for further investigation of PET as a screening tool alongside WBGT.

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INTRODUCTION

More than 800 city marathons and half marathons are arranged around the world. The New York City Marathon has had the largest number of participants, followed by Gothenburg's half-marathon, the latter with over one million runners since its start in 1980. A considerable number of marathon runners suffer from different medical encounters, of which some are so severe that they require ambulance assistance.[1–4] The majority of the collapses requiring ambulance assistance are caused by exertional heat illness, which encompasses exercise associated collapse (most commonly), heat stroke or in extreme cases, cardiac arrest.

Weather is a critical extrinsic risk factor at endurance events.[4 5] A common measure of the likely impact of weather used during sporting events is the Wet-Bulb Globe Temperature (WBGT) index.[6] The WBGT has been used to predict runners' performance and to decide whether specific races should be run or not.[7] It is recommended as a screening tool to assess heat stress during sporting events by the International Olympic Committee and numerous other leading sports federations.[8] Despite its 'brand recognition', the WBGT has limitations and has frequently been criticised in the scientific literature.[9–11] It underestimates heat stress when evaporation of sweat is restricted, for example, at high air humidity or low wind speed. Measurement errors are common due to non-standard measurement devices and unsatisfactory calibration procedures. Modern instruments calculate WBGT from a weighted combination of three sensors: a globe thermometer, a shielded air temperature sensor and a minimally shielded air humidity sensor. A 50 mm globe is now commonly used rather than the standard 150 mm, and the responses of the air humidity and air temperature sensors to solar radiation are not standardised.

Another concern with WBGT is that it is an empirically based index. That is, although WBGT is sensitive to the four most important variables that determine environmental heat stress (air temperature (T_a), solar radiation (G_{rad}), vapour pressure (e_a) and wind speed (w_s)), neither the individual measurement components nor the combined index responds in the same way as does a human body. The index does not approximate a specific physiological parameter, and must be interpreted in relation to physiological studies (eg, sweating) or epidemiological studies (eg, deaths). Further, because the globe temperature and unshielded air humidity measurements each respond to a combination of meteorological variables, the WBGT instrumental setup does not permit a detailed examination of how the individual meteorological variables contribute to thermal stress which consequently may be a limitation in diagnostics.

Other methods to assess heat stress including the Physiological Equivalent Temperature index

Let us consider alternatives for assessing heat stress—the so-called rational human indices including the Physiological Equivalent Temperature

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(PET) index—detailed below. These indices are based on explicit calculations of human energy balance using heat transfer equations and models for the biophysical processes that underpin the thermal state of the human body. The two most commonly used rational indices are the PET index[12] and the Universal Thermal Climate Index (UTCI).[13]

The PET index is based on the Munich Energy-balance model for individuals.[14] It is defined as the air temperature at which, in a typical indoor setting, the heat budget of the human body is balanced with the same core and skin temperature as under the complex outdoor conditions to be assessed.[12] The idea is to transfer actual outdoor thermal conditions to an equivalent indoor environment in which the same thermal sensation is expected.[15] Another index, the UTCI is based on the multi-node 'Fiala' model.[16] It is defined as the air temperature in the reference condition which would elicit the same physiological reaction as predicted by the dynamic response of the physiological model.[17] PET and UTCI model the thermal conditions of the human body in a physiologically relevant way and include all meteorological variables relevant for heat exchange between the body and its environment, that is, T_a, e_a, w_s and mean radiant temperature (T_{mrt}). The latter describes the environmental radiative load on a person[18] and is one of the most important meteorological variables governing the human energy balance and thermal comfort.[15] In addition, they consider the metabolic rate and clothing. Whereas the PET index is calculated for a person with light activity and clothing value of 0.9,[12] the UTCI is calculated for a person with moderate activity and includes a dynamic clothing insulation parameterisation scheme, which dynamically alter the insulation from clothes depending on weather situation.[17] PET and UTCI are commonly used in biometeorological studies worldwide,[19] but are seldom used in sport and exercise medicine.[4 20] Both PET and UTCI can be calculated using standard metrological data that are available for most cites in most countries, along with some basic geographical information using different software, for example, SOLWEIG[21] and Rayman.[22]

To our knowledge, the WBGT index has never been evaluated against rational indices–such as PET or UTCI–using comprehensive ambulance transport data, including medical records. We aimed to compare these indices to assess the best screening tool for predicting health incidents, focusing on collapses requiring ambulance assistance during a major city half marathon.

METHODS

Study area

The Gothenburg half marathon (Göteborgsvarvet) with more than 60 000 registered runners annually was selected for case study. The median age for participants are 41 (men) and 38 (women) and the majority (66%) of the runners are men. The course runs through the central parts of Gothenburg, with a total elevation gain of 44 metres and a cumulative elevation gain of 135 m. The Gothenburg half marathon is organised in mid-May annually.

Gothenburg has a maritime climate with relatively warm winters and cool summers for the latitude. In May, the daily average T_a is 11.6°C and the average high 23.4°C. During high-pressure situations, that is, clear skies and light winds the T_a can reach more than 30°C.

Data

Data concerning the runners' age, gender, physical and medical status along with information regarding all medical encounters resulting in ambulance-required assistances during the Gothenburg half-marathon from 2010 to 2017 was collected from medical records. The number of ambulance transports was used since they represented serious adverse encounters. The ambulance data are reliable and includes information from the first call to the dispatch centre, to locating and loading the patient into the ambulance and finally, the assessment and treatment during transport. Since the number of starters varies between the years, data on number of starters were also collected to assess the rate of ambulance-required assistances and collapses per 1000 starters for each year.

Meteorological data consisted of hourly averages of T_a, w_s, relative humidity (RH) and mean sea-level pressure (MSLP) were collected from the Swedish Meteorological and Hydrological Institute (SMHI) station 71420, located in the city centre of Gothenburg. G_{rad} was collected from the SMHI station 71415 also located in the city centre. Missing T_a, W_s and RH were taken from a station operated by Environment Administration, City of Gothenburg, located near station 71420, and missing MSLP data from SMHI station 72420 (Landvetter Airport), located 20 km inland. MSLP was converted to station pressure assuming a standard atmosphere. These meteorological data were used to calculate the T_{mrt}, WBGT, PET and UTCI.

Thermal comfort indices

As on-site WBGT sensors were not available the WBGT index[6] was calculated by the following equation:

$$WBGT = 0.7_{Tw} + 0.2_{Tg} + 0.1_{Td} \quad (1)$$

where T_w is the natural wet-bulb temperature, T_g is the globe temperature and T_d is the dry-bulb temperature. T_w was calculated according to Liljegren *et al*[23] using the R package *wbgt*.[24] The Liljegren method includes the effects of absorption of both long-wave and short-wave radiation by the wick and has shown to have an accuracy of better-than 1°C across eight test-sites.[23] T_g was estimated according to Okada and Kusaka[25] using T_a, w_s and G_{rad}. The Okada and Kusaka model was used in preference to the Liljegren *et al* model, as the former was calibrated using a standard size 150 mm globe and the latter a 50 mm globe. T_d was assumed to be the same as T_a.

PET and the UTCI were calculated using the SOLWEIG ID software.[21] T_a, RH, w_s and G_{rad} were used as input, whereas the T_{mrt} was calculated by the software. UTCI makes use of w_s at a standard observation height (10 m a.g.l.) whereas PET requires w_s at the centre of gravity for a standing person (1.1 m a.g.l.). Hence w_s was recalculated down from 10 to 1.1. m using a standard power law function[26] with an exponential constant of 0.2. Absorptions coefficients for short-wave and long-wave radiation for the standing person were set to 0.7 and 0.97, respectively. Albedo and emissivity for surrounding surfaces were set to 0.20 and 0.95, respectively.

Predictive methods

Linear regression models were used to assess the association between rates of ambulance-required assistances and collapses and T_a, WBGT, UTCI and PET. The models were evaluated using two approaches: Monte-Carlo simulations and leave-one-out-cross-validation (LOOCV).

CIs for the Pearson correlation, R, can be estimated using the Fisher z-transform provided that: the dataset is bimodally normal, the observations are independent and the sample size is large.[27] Given that the last criterium is not satisfied here, we estimated the 95% CI for R (and hence R^2) using Monte-Carlo simulations. For each model, 10 000 synthetic datasets were created by adding random deviations (with zero mean and the

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SD of the regression residuals) to the regression line at the locations of the observed predictors. The correlation was calculated for each synthetic dataset, and the 95% CI on R ($CI_{95,MC}$) was thus estimated as the range encompassing 95% of the synthetic values.

Additionally, uncertainty was quantified using LOOCV, which successively leaves out one sample from a training data set and estimates models based on the remaining n-1 samples. The LOOCV uncertainty, was quantified by the root-mean-square-error (RMSE) and the range in LOOCV R^2. The former is defined as:

$$RMSE_{LOOCV} = \sqrt{\frac{1}{n}\sum_{i=1}^{n} MSE_i} \quad (2)$$

where n is the number of samples for the regression analysis and MSE is the mean square error. The MSE was calculated using:

$$MSE_i = (y_i - {}_i)^2 \quad (3)$$

where y_i is the observed rate of ambulance-required assistances or collapses for year *i*, and $\hat{y}_i$ is the LOOCV-estimated rate of ambulance-required assistances or collapse for year *i*. The range in R^2 (denoted RR^2_{LOOCV}) is simply the span of the lowest to highest R^2 of the LOOCV models.

Separate Monte-Carlo simulations were used to statistically-compare the relative predictive capacity of the variables T_a, WBGT, UTCI and PET. For each variable *a*, a 'population' was created consisting of 100 000 value-pairs with bivariate normal distribution and Pearson correlation coefficient $R^*_a \approx R_a$, where R_a is the observed correlation of variable *a* versus ambulance-required assistances or collapses (see below for an explanation of R^*). Samples of eight pairs (representing 8 years) were then drawn from this population 100 000 times, and the fraction of samples for which the Pearson correlation coefficient *R* exceeded the R_b (of variable *b* vs ambulance-required assistances or collapses) was used as an estimate of the p value for the hypothesis $p=\Pr(R>R_b \mid R_a)$. Results are presented as the confidence (1—p) in an alternative hypothesis, denoted as $R_b \geq R_a$.

Note that the median correlation coefficient ($\tilde{R}$) of samples drawn from a population with R_a is an overestimate of R_a. For example, in a case of where the population had $R=0.825$, the median (calculated from 2×10^6 draws of 8 pairs) was $\tilde{R} \approx 0.849$. This in turn yields a biased probability estimate $\Pr(R \geq R_a \mid R_a) \approx 0.57 \neq 0.5$. We have attempted to account for this bias when calculating p values. No guidance on how to construct a population to yield unbiased $\tilde{R}$ could be found in the literature, but trial-and-error suggested that drawing from a population with correlation coefficient

$$R^*_a = R_a - \frac{\sqrt{2}}{(n+1)^2} \quad (4)$$

where there are n=8 pairs in each yielded samples for which $\tilde{R} \approx R_a$ over the range of R_a of this study. Continuing the above example, the median of samples drawn from a population with R^*_a was $\tilde{R} = 0.829$, much closer to the original $R_a=0.825$.

The correlation ratio (η), a measure of the dispersion in a categorised continuous variable, was calculated in order to analyse how well the PET stages of physiological stress[28] and the WBGT generalised heat stress flag colour warnings for runners and risk for hyperthermia[29] match the medical incident rates. A ratio η=0 implies all categories have the same mean, whereas η=1 implies all categories have a unique value.

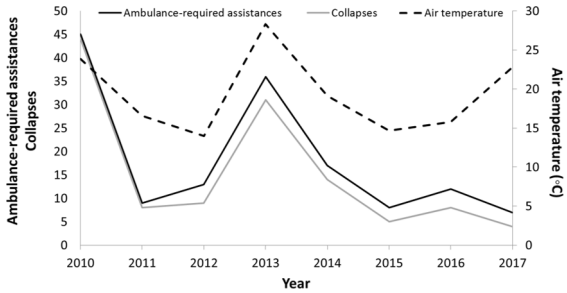

Figure 1 Absolute number of ambulance-required assistance, collapses and maximum air temperature during the Gothenburg half marathon (2010–2017). Modified after Carlström *et al*.[4]

RESULTS

Ambulance required assistances and collapses

Between 2010 and 2017, 147 runners of the 357 896 participants in the Gothenburg half marathon or 0.42 per 1000 starting runners required ambulance assistance. A majority (84%) of these were due to exercise associate collapse. The mean age of those who required ambulance assistance was 34 years (range 17–66; SD=12), which is less than the average age of all participants. A majority (61%) were men, but when adjusted for the lower ratio of female runners, men were slightly underrepresented.

Yearly distribution of runners requiring ambulance assistance and collapses

The number of ambulance-required assistances and collapses varied between the years (figure 1). High number of collapses (>30) were observed at T_a >24°C whereas low number of collapses (<10) were observed at T_a <24°C, indicating that the warmer weather, the larger the number of collapses requiring ambulance assistance.

The influence of weather

T_a alone explained 46% of the variance in rate of ambulance-required assistances (p=0.06) and 44% of the variance in rate of collapses (p=0.075) (table 1). The rational indices, which combine all metrological variables influencing the level of heat stress, explained an even larger fraction of the variance: PET explained 72% of the variance in rate of ambulance-required assistances (p=0.008) and 71% of the variance in rate of collapses (p=0.008), followed by the UTCI (64%, p=0.02 and 64%, p=0.02). This is to be compared with the empirical WBGT index, which explained 56% of the variance in rate of ambulance-required assistances (p=0.03) and 56% of the variance in rate of collapses (p=0.03). The forward stepwise multiple regression-model selected only PET, which implies that using multiple indices will not significantly improve the model fit.

All models were evaluated by means of LOOCV. The PET and UTCI models showed the smallest $RMSE_{LOOCV}$ of all models (table 1).

The highest confidence for both ambulance-required assistances and collapses was for the hypothesis $R_{PET}>R_{Ta}$, with confidence 0.83 and 0.84, respectively (table 2). Experimentation suggested that around 17 years would be required to attain a confidence of 0.95 for $R_{PET}>R_{Ta,}$ and around 40 years would be required to attain a confidence of 0.95 for $R_{PET}>R_{WBGT}$, given the correlations vs ambulance-required assistances reported in this study.

The rate of collapses increased with higher PET by 0.04±0.03 per 1000 participants for every °C increase in PET (figure 2).

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Table 1 Coefficient of determination (R^2) between maximum air temperature (T_a), maximum WBGT, PET and UTCI (independent variables) and rates of ambulance-required assistances and collapses (depended variables) over 2010–2017 (n=8) in the Gothenburg's half marathon; p value estimated using the Fisher z-transform; 95% CI for R^2 derived from Monte-Carlo simulations ($CI_{95,MC}$); the range of R^2 estimated by leave-one-out-cross-validation (RR^2_{LOOCV}); and the root-mean-square-error from LOOCV ($RMSE_{LOOCV}$)

Dependent variable	Independent variable	R^2	P value	$CI_{95,MC}$	RR^2_{LOOCV}	$RMSE_{LOOCV}$ (per 1000 starting runners)
Ambulance-required assistances	T_a	0.46	0.063	0.04 to 0.88	0.37–0.69	0.31
	WBGT	0.56	0.031*	0.15 to 0.90	0.45–0.76	0.3
	UTCI	0.64	0.017*	0.27 to 0.92	0.52–0.76	0.28
	PET	0.72	0.008†	0.41 to 0.94	0.59–0.85	0.25
Collapses	T_a	0.44	0.075	0.03 to 0.86	0.38–0.64	0.32
	WBGT	0.56	0.033*	0.13 to 0.90	0.46–0.76	0.31
	UTCI	0.64	0.017*	0.27 to 0.92	0.56–0.76	0.29
	PET	0.71	0.008†	0.40 to 0.94	0.62–0.83	0.26

*Significant at 95% level.
†Significant at 99% level.
PET, Physiological Equivalent Temperature; UTCI, Universal Thermal Climate Index; WBGT, Wet-Bulb Globe Temperature.

However, the rate of collapses was almost two times higher in 2010 (1.1 per 1000 starting runners) than in 2013 (0.66 per 1000 starting runners) although the maximum PET was almost as high (37.5°C and 36.4°C, respectively). In both these years, the Gothenburg half marathon took place during the hitherto hottest day of the year. In 2010, the race was preceded by a change in weather, with a large increase in maximum PET from 26°C on the day before to 37.5°C on the day of the race. In 2013, however, the maximum PET was more or less the same the day before the race (35.7°C) as on the day of the race (36.4°C).

Level of heat stress

The maximum PET during the races ranged between 19°C and 38°C, which corresponds to the range of 'no thermal stress' to 'strong heat stress'.[28] Corresponding maximum WBGT ranged between 16°C and 27°C, which corresponds to the range of 'GREEN' to 'RED' flag colour, hat is, low to high risk for hyperthermia.[29] High rate of collapses (>0.6 per 1000 starting runners) was observed during years with 'strong heat stress' and 'RED' flag colour; whereas low rates of collapses (<0.3 per starting 1000 runners) were in general observed during 'no thermal stress' or 'slight heat stress' and 'GREEN' or 'YELLOW' flag colour, respectively (table 3).

Table 2 Confidence estimates (1—p values) for the hypothesis $R_b > R_a$ for Pearson correlation coefficients for the four independent variables T_a, WBGT, UTCI and PET versus ambulance-required assistances (above) and versus collapses (below)

Dependent variable	Independent variable (R_a)	Independent variable (R_b)			
		T_a	WBGT	UTCI	PET
Ambulance-required assistances	T_a	0.48	0.62	0.72	0.83
	WBGT	0.35	0.50	0.61	0.74
	UTCI	0.26	0.39	0.50	0.64
	PET	0.17	0.28	0.39	0.52
Collapses	T_a	0.48	0.65	0.76	0.84
	WBGT	0.33	0.49	0.62	0.74
	UTCI	0.23	0.38	0.50	0.63
	PET	0.15	0.27	0.40	0.52

Diagonal elements (representing confidence for $R_a > R_a$ which is by definition 0.5) give an indication of the uncertainty in the Monte-Carlo estimation procedure.
PET, Physiological Equivalent Temperature; UTCI, Universal Thermal Climate Index; WBGT, Wet-Bulb Globe Temperature.

Although the number of races in each category was very small, the correlation ratio of collapses (η=0.93) and ambulance-required assistances (η=0.94) was higher for PET stages of heat stress than for WBGT generalised heat stress flag colour (η=0.63, and 0.64 for collapses and ambulance-required assistances, respectively). This means that the PET stages of heat stress stratified the rate of collapses better that the WBGT risk categories for hyperthermia.

Compared with the PET, the WBGT index underestimated heat stress and the underestimation was accentuated at higher PET (figure 3 left), that is, higher grades of physiological stress (R^2=0.89, p=0.000 and 0.88, p<0.001). T_{mrt} explained 82% (p=0.002) of the difference between the WBGT and PET and the underestimation of WBGT was accentuated by higher T_{mrt} (figure 3 right), that is, higher radiant heat load. Neither T_a, e_a nor w_s were significantly correlated with the difference between WBGT and PET. The UTCI also underestimated heat stress compared with PET, however, to less extent than the WBGT.

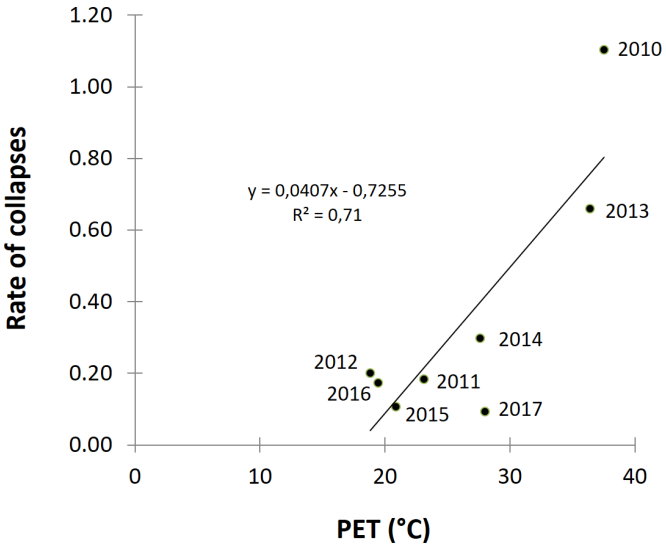

Figure 2 Rate of collapses (per 1000 starting runners) versus maximum Physiological Equivalent Temperature (PET).

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Table 3 Rate of collapses and ambulance-required assistances (per 1000 starting runners); maximum PET during the race (max PET) and associated stage of physiological stress[28]; maximum WBGT during the race and associated FLAG colour and risk for hyperthermia[29] between 2010 and 2017, in descending order

Collapses	Ambulance-required assistances	Max PET (°C)	Max PET stress	Max WBGT (°C)	Max WBGT flag colour (risk for hyperthermia)	Year
1.10	1.13	37.5	Strong heat stress	27	Red (high)	2010
0.66	0.77	36.4	Strong heat stress	27.2	Red (high)	2013
0.30	0.36	27.6	Slight heat stress	21.8	Yellow (moderate)	2014
0.20	0.29	18.8	No thermal stress	16.2	Green (low)	2012
0.18	0.21	23.1	Slight heat stress	20.4	Yellow (moderate)	2011
0.17	0.26	19.5	No thermal stress	19.9	Yellow (moderate)	2016
0.11	0.17	20.9	No thermal stress	17.6	Green (low)	2015
0.09	0.16	28.0	Slight heat stress	23.8	Red (high)	2017

PET, Physiological Equivalent Temperature; WBGT, Wet-Bulb Globe Temperature.

DISCUSSION

We aimed to compare the different thermal indices' ability to predict rates of incidents requiring ambulance assistance and explain the variance in rates of ambulance-required assistances and collapses between years. The PET index had the best performance and the strongest correlation with both rates of ambulance-required assistances and collapses, followed by the UTCI, whereas the WBGT index showed substantially poorer correlation and ability to predict rates of ambulance-required assistances and collapses (table 1).

PET stages of thermal stress also matched the medical incident rates better than the WBGT generalised heat stress flag colour warnings for runners and risk for hyperthermia (table 3). The apparent discrepancy between WBGT flag colour and rates of medical emergencies for year 2017, however, deserves closer inspection. In this paper, heat stress for a race was characterised by the maximum index value that occurred during the race. The weather on the day of the race in 2017 was noteworthy for the cold change that began soon after the race started. Thus, the maximum index value at the beginning of the race did not represent the heat stress conditions during a large part of the race, especially in the latter part of the race. Additionally, the WBGT index maximum in 2017 (23.8°C) was at the lower-end of the RED flag range, while the PET maximum (28.0°C) was at the upper end of the 'slight heat stress' range.

The rate of collapses increased by 0.04±0.03 per 1000 participants for every °C increase in PET (figure 2). This implies that for the Gothenburg half-marathon with about 45 000 participants annually the number of collapses increases with 1.8±1.4 runners for every °C increase in PET. However it seems that large increase in PET (>10°C) from the day before as in 2010 may have an impact on the rate of collapses. A possibility explanation for the increase in rate is that the runners have not acclimatised to the hotter conditions as it takes days for the body to do so.[30] Another possible explanation is that the runners did not expect such hot conditions and thus did not drink enough before, during and after the race, wear unsuitable cloths (ie, dark-coloured and too much clothes) and/or did run too fast.

Compared with the PET, the WBGT underestimated heat stress, the underestimation being accentuated by higher PET and T_{mrt}, that is, higher heat stress and radiant heat load (figure 3). High T_{mrt} occurs during warm and sunny weather conditions,[31] when it is the most important meteorological variable governing the human energy balance and thermal comfort (heat load).[15] The fact that the WBGT underestimates heat stress during conditions of high radiant heat load and high physiological stress can explain its poorer correlation with ambulance-required assistances and collapses. Another possible factor to the poorer correlation could be that the T_g needed for the WBGT calculations was not measured, but

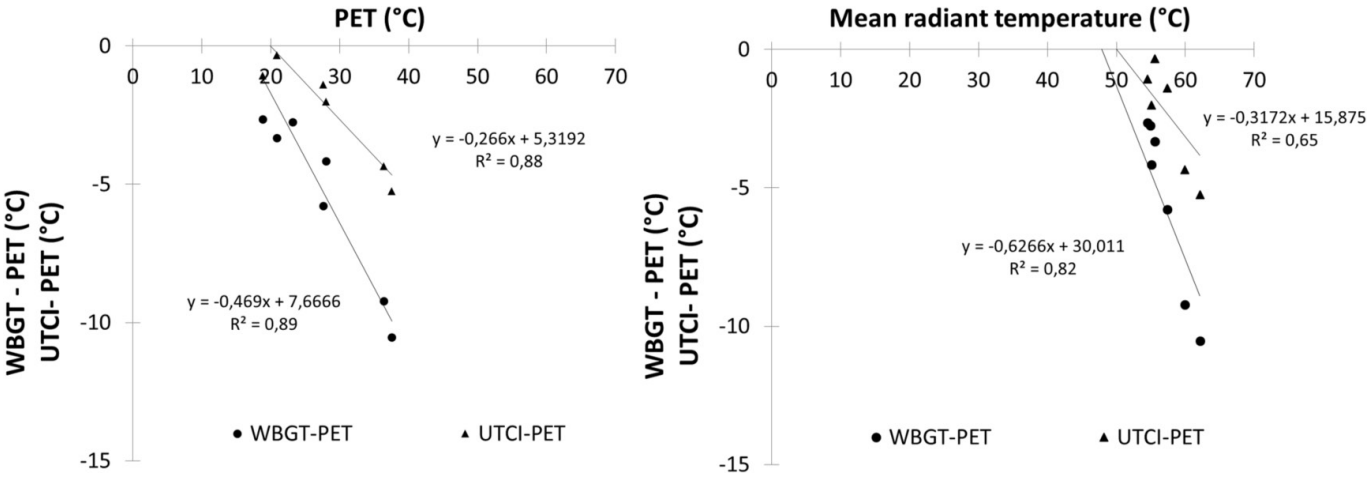

Figure 3 Difference between maximum Physiological Equivalent Temperature (PET) and maximum Universal Thermal Climate Index (UTCI) and maximum Wet-Bulb Globe Temperature (WBGT) as functions of maximum PET (left) and maximum mean radiant temperature (T_{mrt}) (right).

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calculated using Okada and Kusaka.[25] According to Budd[9] the omission of T_g measurements may cause errors, and differences caused by the omission of T_g measurements can thus not be excluded.

Even though PET and UTCI are not designed for athletes and may in fact underestimate heat stress as they assume light and moderate activity, respectively, the comparison of models presented here suggest that rational indices are better screening tools for the assessment of heat-stress hazard during the Gothenburg half-marathon, as compared with the WBGT index. The fact that rational indices explain 64%–72% of the variance in ambulance-required assistances and collapses and that the PET stages of heat stress match the rate of collapses better that the WBGT generalised heat stress flag colour and risk categories for hyperthermia suggests that it might not only be possible to predict the number of required ambulance transports and collapses with a high accuracy, but that there may be good substitute indices for the criticised WBGT.

Our study may have important clinical implications, adding to the knowledge of thermal stressors' role for medical encounters at endurance competitions, but also for planning f medical resources for races. More studies using measurements of T_g instead of estimations from T_a, w_s and G_{rad} are needed, as are more studies from other marathons in other climates. Once these data are gathered, an appropriate group could consider whether the WBGT should be replaced.

CONCLUSIONS

Although the PET was not designed for athletes, it explained a larger fraction of the variance in rates of ambulance-required assistances and collapses than the commonly used WBGT index. The PET stages of heat stress also matched the medical incident rates better that the WBGT generalised heat stress flag colour and risk categories for hyperthermia. Compared with the PET, the WBGT index underestimated heat stress, especially at high radiant heat load, where heat indices may be most needed in clinical practice. The rates of ambulance-required assistances and collapses increased with increasing level of heat stress however large increase in heat stress from the day before seems to have an impact on the rate of collapses. Although more data are needed, we conclude that the PET is a better predictor of ambulance-required assistances and collapses than the WBGT index.

Summary box

What are the findings?

- The rational indice Physiological Equivalent Temperature (PET) and Universal Thermal Climate Index (UTCI) showed stronger correlations with ambulance-required assistances and collapses than the Wet-Bulb Globe Temperature (WBGT) index.
- PET stages of thermal stress match the rates of collapses better that the WBGT generalised heat stress flag colour warnings for runners and risk for hyperthermia.
- Compared with the PET, the WBGT underestimates heat stress, especially at high radiant heat load.
- Large increase in heat stress the day before the race seems to have an impact on the rate of collapses.

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Data availability statement Data may be obtained from a third party and are not publicly available. The data are based on ambulance patient records and registers from the Gothenburg half marathon organisation. According to the ethical permission, deidentified data can be shared to other researchers involved in a linked study e.g. for comparative reasons. The data can be obtained from the first author, ST or EC (coauthor). There is no time limit on the availability of the data.

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